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物理数学 Report

$f(z)$：整関数　べき級数に展開可能

全複素平面で解析的　　$e^z = 1 + z + \frac{z^2}{2} + \cdots$,　$\frac{1}{z}$

($z=\infty$ は含まない)

$$f(z) = \sum_{n=0}^{\infty} f_n z^n = \overbrace{\sum_{n=0}^{N-1} f_n z^n}^{f_N(z)} + \overbrace{\sum_{n=N}^{\infty} f_n z^n}^{R_N(z)}$$

$$\|f_{N-1}\|^2 = \sum_{n=0}^{N-1} n! |f_n|^2 \longrightarrow \begin{cases} \infty \\ \text{finite} \end{cases} \qquad \|f_{N-1}\| \le \|f_{N-2}\| \le \|f_{N=0}\| \le \cdots$$

$\mathcal{F} = \{ f(z) : \text{整関数 かつ } \|f\| < \infty \} = \overline{\{1, z, z^2, z^3, \cdots\}}$

$$(f,g) = \int d\mu(z) \sum_{n=0}^{\infty} f_n^* (z^*)^n \sum_{m=0}^{\infty} g_m z^m = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} \int d\mu(z)\, f_n^* (z^*)^n g_m z^m = \sum_{n=0}^{\infty} n!\, f_n^* g_n$$

$$\left| \int d\mu(z) \sum_{n=N}^{\infty} f_n^* (z^*)^n \sum_{m=N}^{\infty} g_m z^m \right|^2 = |(R_N^f, R_N^g)|^2 \le \|R_N^f\|^2 \|R_N^g\|^2$$

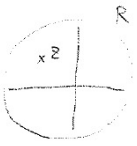
$$\sum_{n=N}^{\infty} n! |f_n|^2 \longrightarrow 0 \quad (N \to \infty)$$

(16)　$e_a(z) = e^{a^* z} = \sum_{n=0}^{\infty} \frac{(a^* z)^n}{n!} = \sum_{n=0}^{\infty} U_n(a)^* U_n(z)$　　$\left(U_n(z) = \frac{z^n}{\sqrt{n!}} \right)$

$$(e_a, f) = \sum_{n=0}^{\infty} U_n(a) \underbrace{(U_n, f)}_{c_n} \qquad f \in \mathcal{F} \Rightarrow f(z) = \sum_{n=0}^{\infty} c_n U_n(z)$$

$$= \sum_{n=0}^{\infty} c_n U_n(a) \qquad (U_n, f) = c_n$$

$$= f(a) \qquad f(z) = (e_z, f)$$



$$|f(z)| = |(e_z, f)| \le \|e_z\| \|f\| = e^{\frac{|z|^2}{2}} \|f\| \le e^{\frac{R^2}{2}} \|f\|$$

〔問2〕完備性の証明

$f_n(z) \in \mathcal{F}$

$\|f_n - f_m\| \to 0$ for $n, m \to \infty$ とする　(19)を使って

$|f_n(z) - f_m(z)| \le \|f_n - f_m\| e^{\frac{|z|^2}{2}} \le \|f_n - f_m\| e^{\frac{R^2}{2}} \to 0 \quad (n, m \to \infty)$

if $|z| \le R$ (z に関係なく)

各 z ごとに $f_n(z) \longrightarrow {}^{\exists} f(z) \in \mathbb{C}$ かつ一様収束

$\to f_n(z)$ が $|z| \le R$ で解析的 $\to f(z)$ も $|z| \le R$ で解析的

R は任意より $f(z)$ は全平面で解析的

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$\left(\begin{array}{l} \forall \varepsilon > 0 \ \exists N, \ n \geq N \quad \|f - f_n\| < \varepsilon \\ \text{十分大きな } m \text{ に対して } \|f - f_m\| \leq \varepsilon N_n \end{array}\right)$

$f = \sum_{n=0}^{\infty} c_n U_n$

$(zf, g) = (f, \frac{\partial}{\partial z} g)$ を $f = U_n$, $g = U_m$ について証明すれば十分である。

$(z \cdot z^n, z^m) = m! \, \delta_{n+1,m}$

$\left(z^n, \frac{\partial}{\partial z} z^m\right) = (m-1)(z^n, z^{m-1})$

$= (m-1)(m-1)! \, \delta_{n,m-1}$

$= m! \, \delta_{n+1,m}$

$e^{-ax} f(z) e^{ax} = e^{-ax} \sum_{n=0}^{\infty} f_n z^n e^{ax} = \sum_{n=0}^{\infty} f_n e^{-ax} z e^{ax} e^{-ax} z e^{ax} \cdots e^{-ax} z e^{ax}$

$= \sum_{n=0}^{\infty} f_n (e^{-ax} z e^{ax})^n = f(e^{-ax} z e^{ax})$

$\left[z \frac{\partial}{\partial z}, z\right] = z$

$[AB, C] = [A, C]B + A[B, C]$

$[A, A_2 \cdots A_n, C] = [A, C] A_2 \cdots A_n + A_1 [A_2, C] A_3 \cdots A_n + \cdots + A_1 \cdots A_{n-1} [A_n, C]$

$e^{-\alpha z \frac{\partial}{\partial z}} z e^{\alpha z \frac{\partial}{\partial z}} = z - \frac{\alpha}{1!}\left[z \frac{\partial}{\partial z}, z\right] + \frac{\alpha^2}{2!}\left[z \frac{\partial}{\partial z}, \left[z \frac{\partial}{\partial z}, z\right]\right] + \cdots$

$= \left(1 - \frac{\alpha}{1!} + \frac{\alpha^2}{2!} - \cdots\right) z = e^{-\alpha} z$

$e^{-\alpha z \frac{\partial}{\partial z}} f(z) e^{\alpha z \frac{\partial}{\partial z}} = f(e^{-\alpha} z)$

$z \frac{\partial}{\partial z} f(z) = n f(z)$

$f(cz) = c^n f(z)$

c で微分する

$z f'(cz) = n c^{n-1} f(z)$

$c = 1$ とおく

$z f'(z) = n f(z)$

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(26)

$$\langle \xi | a | 0 \rangle = \frac{1}{\sqrt{2}} \left(\xi + \frac{\partial}{\partial \xi} \right) \langle \xi | 0 \rangle \qquad \text{“} X_0(\xi) \text{”}$$

$$-\frac{1}{2}\left(\frac{\partial}{\partial \xi}\right)^2 + \frac{\xi^2}{2} = a^\dagger a + \frac{1}{2}$$

$$a a^\dagger + a^\dagger a = -\left(\frac{\partial}{\partial \xi}\right)^2 + \frac{\xi^2}{2} = 2 a^\dagger a + 1$$

$$\left[-\frac{1}{2}\left(\frac{\partial}{\partial \xi}\right)^2 + \frac{\xi^2}{2} \right] \langle \xi | n \rangle = \left(n + \frac{1}{2} \right) \langle \xi | n \rangle \qquad \text{“} X_n(\xi) \text{”}$$

$$a^\dagger a | n \rangle = n | n \rangle$$

$$\langle \xi | a^\dagger a | n \rangle = n \langle \xi | n \rangle$$

$$\left(-\frac{1}{2}\left(\frac{\partial}{\partial \xi}\right)^2 + \frac{\xi^2}{2} \right) \langle \xi | n \rangle = n \langle \xi | n \rangle$$

$$g(\xi) = \langle \xi | g \rangle$$

$$\int d\mu(\xi)\, f^*(\xi)\, g(\xi) = \int d\mu(\xi) \langle f | \xi \rangle \langle \xi | g \rangle = \sum_{n=0}^{\infty} \tilde{f}_n^*\, \tilde{g}_n = \sum_{n=0}^{\infty} \langle f | n \rangle \langle n | g \rangle$$

$$f(\xi) = \sum_{n=0}^{\infty} \tilde{f}_n U_n(\xi)$$

$$\int d\mu(\xi)\, |\xi\rangle\langle\xi| = \sum_{n=0}^{\infty} |n\rangle\langle n| = 1$$

$$= \langle \xi | f \rangle = \sum_{n=0}^{\infty} \langle \xi | n \rangle \langle n | f \rangle$$

$$= \sum_{n=0}^{\infty} U_n(\xi)\, \tilde{f}_n$$

$$e^A e^B = e^{A + B + \frac{1}{2}[A,B] + \circledast [A,[A,B]] + \circledast [[A,B],B] + \cdots} = e^C$$

$$C = \log e^C = \log e^A e^B = \log\left(1 + (e^A e^B - 1)\right) \qquad \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots$$

$$= (e^A e^B - 1) - \frac{1}{2}(e^A e^B - 1)^2 + \cdots$$

$$e^A e^B - 1 = \left(1 + A + \frac{A^2}{2!} + \cdots\right)\left(1 + B + \frac{B^2}{2!} + \cdots\right) - 1$$

$$= (A + B) + \frac{1}{2!}(A^2 + 2AB + B^2) + \frac{1}{3!}(\cdots)$$